

SFR Preliminary Analysis

Sander Elliott

2026-04-23

Packages

```
library(tidyr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(glatos)
```

```
## version 0.9.0 ('pretty-fragrant-rye')
```

```
library(sf)
```

```
## Linking to GEOS 3.13.1, GDAL 3.11.4, PROJ 9.7.0; sf_use_s2() is TRUE
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats   1.0.1      v readr     2.1.6
## v ggplot2   4.0.1      v stringr  1.6.0
## v lubridate 1.9.4      v tibble   3.3.1
## v purrr     1.2.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()    masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(dataRetrieval)
```

```
## Warning: package 'dataRetrieval' was built under R version 4.5.3
```

```
library(rosm)
```

```
## Warning: package 'rosm' was built under R version 4.5.3
```

```
library(raster)
```

```
## Loading required package: sp
##
## Attaching package: 'raster'
##
## The following object is masked from 'package:dplyr':
##
##     select
```

```
library(brms)
```

```
## Warning: package 'brms' was built under R version 4.5.3
```

```
## Loading required package: Rcpp
## Loading 'brms' package (version 2.23.0). Useful instructions
## can be found by typing help('brms'). A more detailed introduction
## to the package is available through vignette('brms_overview').
##
## Attaching package: 'brms'
##
## The following object is masked from 'package:stats':
##
##     ar
```

Data Prep

Raw Data

```
SNSbst <- read.csv("data/SNSbst_allyrs.csv")
SNSfsh <- read.csv("data/SNSfsh_allyrs.csv")
glimpse(SNSbst) ## Detection history
```

```
## Rows: 114,631
## Columns: 26
## $ Period      <chr> "1997-07-14 14:15:00", "1998-09-03 13:08:00", "1997-0~
## $ tagtype     <chr> "Acoustic", "Acoustic", "Acoustic", "Acoustic", "Acou~
## $ Species     <chr> "SNS", "SNS", "SNS", "SNS", "SNS", "SNS", "SNS", "SNS~
## $ FishID      <chr> "1997-SNS-0004-DMR", "1997-SNS-0004-DMR", "1997-SNS-0~
```

```
## $ TagId          <chr> "2336", "2336", "2336", "2336", "16", "16", "16", "16~
## $ Event          <chr> "No estimated end of tag life", "No estimated end of ~
## $ pings          <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, 1~
## $ FirstTS        <chr> "", "", "1997-07-14 14:15:00", "1998-09-03 13:08:00",~
## $ LastTS         <chr> "", "", "1997-07-14 14:15:00", "1998-09-03 13:08:00",~
## $ SensorType     <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ avgSensorValue <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, 0~
## $ avgComputedValue <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, 0~
## $ Codespace      <chr> "2336", "2336", "2336", "2336", "16", "16", "16", "16~
## $ IDCode         <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, 30385~
## $ Frequency      <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ avgPower       <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ AntennaID      <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ RxID           <chr> "", "", "", "", "", "", "", "", "", "", "", "", "", "", "~
## $ DeploymentUID  <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, 7~
## $ Easting        <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, 4~
## $ Northing       <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, 4~
## $ LocationCode   <chr> "", "", "45B", "45D", "", "", "45D", "46", "", "", "4~
## $ RKM            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, 4~
## $ UTMZone        <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, 1~
## $ SiteCode       <chr> "", "", "", "", "", "", "", "", "", "", "", "", "", "", "~
## $ AltFishID      <chr> "", "", "", "", "", "", "", "", "", "", "", "", "", "", "~
```

`glimpse(SNSfsh) ## Fish information`

```
## Rows: 271
## Columns: 39
## $ FishID          <chr> "1997-SNS-0004-DMR", "1997-SNS-0004-DMR", "1998-SNS~
## $ Species         <chr> "SNS", "SNS", "SNS", "SNS", "SNS", "SNS", "SNS", "S~
## $ TagID           <chr> "2336", "2336", "16", "16", "22", "22", "A69-9001-3~
## $ TagType         <chr> "Acoustic", "Acoustic", "Acoustic", "Acoustic", "Ac~
## $ Origin          <chr> "", "", "", "", "", "", "", "", "", "", "", "", "", "", "~
## $ FishIDs         <chr> "", "", "", "", "", "", "", "", "", "", "", "", "", "", "~
## $ ProjectID       <chr> "", "", "", "", "", "", "", "", "", "", "", "", "", "", "~
## $ ReleaseSite     <chr> "", "", "", "", "", "", "", "", "", "", "", "", "", "", "~
## $ CaptureDate     <chr> "1997-07-14 14:15:00", "1998-09-03 13:08:00", "1998~
## $ CaptureLocation <chr> "45B", "45D", "45D", "46", "45A", "46", "44", "13",~
## $ Easting         <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, 434~
## $ Northing        <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, 485~
## $ ForkLength      <dbl> 93.0, 96.2, 85.5, 84.1, 81.5, 82.0, 76.8, 76.5, 83.~
## $ TotalLength     <dbl> 106.0, 105.5, 96.0, 92.4, 92.0, 92.0, 88.0, 88.3, 9~
## $ LengthUnit      <chr> "cm", "cm", "cm", "cm", "cm", "cm", "cm", "cm", "cm", "cm~
## $ Mass            <dbl> NA, NA, NA, 4500, 6000, 5500, 3700, 3000, 4900, 290~
## $ MassUnit        <chr> "", "", "", "g", "g", "g", "g", "g", "g", "g", "g", "~
## $ Alive           <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ Recapture       <int> 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ InitialRelease  <int> 1, 0, 1, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ Sex             <chr> "", "", "", "", "", "", "F", "", "", "", "", "", "", "~
## $ Interorbital    <dbl> 74.4, NA, 61.9, 58.8, 58.9, 59.2, 61.0, 56.0, 61.0,~
## $ InsideMouth     <dbl> 45.0, NA, 62.9, 46.9, 46.7, 43.1, 46.0, 39.0, 49.0,~
## $ OutsideMouth    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ DorsalScore     <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ CaudalScore     <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ FinClipObs      <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
```

```
## $ Surgeon      <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ VIE          <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ DevelopmentalStage <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ StockingMark1 <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ StockingMark2 <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ FinClip      <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ ScaleSample  <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ SensorType   <chr> "", "", "", "", "", "", "", "", "", "", "", "", "", "", "",~
## $ SensorIDCode <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ SensorCodeSpace <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ System       <chr> "Kennebec", "Kennebec", "Kennebec", "Kennebec", "Ke~
## $ Notes        <chr> "", "", "PIT & carlin match but smaller on recaptur~
```

Detection History

```
## Remove columns for types of tags not used in analysis
## Only keep telemetry detections
SNSdetcln <- SNShst %>%
  filter(Event == "Detection") %>%
  dplyr::select(-Event, -Species, -tagtype, -SensorType, -avgSensorValue, -avgComputedValue, -Frequency,
    -LocationCode, -AntennaID, -AltFishID)
```

Prep Glatos - Convert coordiantes and time

```
detglat_sf <- SNSdetcln %>%
  mutate(
    detection_timestamp_utc = as.POSIXct>LastTS),
    transmitter_id = as.character>IDCode),
    transmitter_codespace = as.character>TagId),
    receiver_sn = RxID,
    animal_id = FishID
  ) %>%
  filter(!is.na(Easting)) %>%
  st_as_sf(coords = c("Easting", "Northing"), crs = 26919) %>% # UTM 19N
  st_transform(4326) %>% # WGS84 lat/long
  mutate(
    deploy_long = st_coordinates(.)[,1],
    deploy_lat = st_coordinates(.)[,2]
  ) %>%
  filter(deploy_lat < 45.1) ## some false detections way north

glat_sf <- detglat_sf %>%
  mutate(array = case_when(deploy_lat < 44.49 & deploy_long > -69.2 ~
    "penob_bay",
    deploy_lat > 44.49 & deploy_long > -69.2 ~
    "penob_riv",
    deploy_long < -69.2 ~
    "ken_r"))
```

```
detglat_sf_3857a <- st_transform(glat_sf, 3857)
```

```
glimpse(glat_sf)
```

```
## Rows: 114,163
## Columns: 22
## $ Period          <chr> "2011-11-14 00:00:00", "2011-11-15 00:00:00", ~
## $ FishID          <chr> "2010-SNS-0002-DMR", "2010-SNS-0002-DMR", "201~
## $ TagID           <chr> "A69-9001-30385", "A69-9001-30385", "A69-9001--
## $ pings           <int> 161, 180, 140, 44, 35, 77, 32, 8, 128, 138, 47~
## $ FirstTS         <chr> "2011-11-14 18:47:41", "2011-11-15 16:02:03", ~
## $ LastTS          <chr> "2011-11-14 23:12:57", "2011-11-15 21:02:22", ~
## $ Codespace       <chr> "A69-9001", "A69-9001", "A69-9001", "A69-9001"~
## $ IDCode          <int> 30385, 30385, 30385, 30385, 30385, 30385, 3038~
## $ RxID            <chr> "104004", "104015", "104004", "104004", "10401~
## $ DeploymentUID    <int> 7383, 7386, 7383, 7383, 7410, 7411, 7412, 7414~
## $ RKM             <dbl> 42.0, 59.0, 42.0, 42.0, 48.0, 55.0, 59.0, 0.0,~
## $ UTMZone         <dbl> 19, 19, 19, 19, 19, 19, 19, 19, 19, 19, 19, 19~
## $ SiteCode        <chr> "12 Gleasons", "9 Rolling Dam", "12 Gleasons",~
## $ detection_timestamp_utc <dtm> 2011-11-14, 2011-11-15, 2011-11-16, 2011-11-1~
## $ transmitter_id   <chr> "30385", "30385", "30385", "30385", "30385", "~
## $ transmitter_codespace <chr> "A69-9001-30385", "A69-9001-30385", "A69-9001--
## $ receiver_sn      <chr> "104004", "104015", "104004", "104004", "10401~
## $ animal_id        <chr> "2010-SNS-0002-DMR", "2010-SNS-0002-DMR", "201~
## $ geometry         <POINT [°]> POINT (-69.78478 44.06664), POINT (-69.7~
## $ deploy_long       <dbl> -69.78478, -69.75957, -69.78478, -69.78478, -6~
## $ deploy_lat        <dbl> 44.06664, 44.19696, 44.06664, 44.06664, 44.124~
## $ array            <chr> "ken_r", "ken_r", "ken_r", "ken_r", "ken_r", "~
```

Use GLATOS

```
glat_events <- detection_events(glat_sf, location_col = "array", condense = TRUE) %>%
  mutate(m_leave = month(last_detection),
         y_leave = year(last_detection),
         d_leave = yday(last_detection),
         m_leave = month(last_detection),
         y_arrive = year(first_detection),
         d_arrive = yday(first_detection))
```

```
## The event filter distilled 114163 detections down to 790 distinct detection events.
```

```
glimpse(glat_events)
```

```
## Rows: 790
## Columns: 14
## $ event          <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16,~
## $ animal_id      <chr> "2010-SNS-0002-DMR", "2011-SNS-0001-DMR", "2011-SNS-00~
## $ location       <chr> "ken_r", "ken_r", "ken_r", "ken_r", "ken_r", "ken_r", ~
## $ mean_latitude  <dbl> 44.05311, 44.13633, 44.18116, 44.18842, 44.15661, 43.9~
## $ mean_longitude <dbl> -69.78126, -69.77063, -69.78113, -69.77812, -69.80353,~
```

```
## $ first_detection <dtm> 2011-11-14, 2011-11-04, 2011-11-09, 2011-11-15, 2012-~
## $ last_detection <dtm> 2012-11-03, 2016-09-12, 2016-10-22, 2017-11-03, 2017-~
## $ num_detections <int> 47, 251, 411, 563, 174, 161, 538, 7, 669, 576, 93, 204~
## $ res_time_sec <dbl> 30668400, 153273600, 156294000, 188348400, 148608000, ~
## $ m_leave <dbl> 11, 9, 10, 11, 4, 11, 7, 9, 8, 6, 5, 10, 8, 10, 7, 5, ~
## $ y_leave <dbl> 2012, 2016, 2016, 2017, 2017, 2019, 2019, 2012, 2019, ~
## $ d_leave <dbl> 308, 256, 296, 307, 113, 311, 211, 247, 214, 181, 146, ~
## $ y_arrive <dbl> 2011, 2011, 2011, 2011, 2012, 2013, 2012, 2012, 2012, ~
## $ d_arrive <dbl> 318, 308, 313, 319, 220, 103, 214, 222, 294, 223, 222, ~
```

Plot Arrays

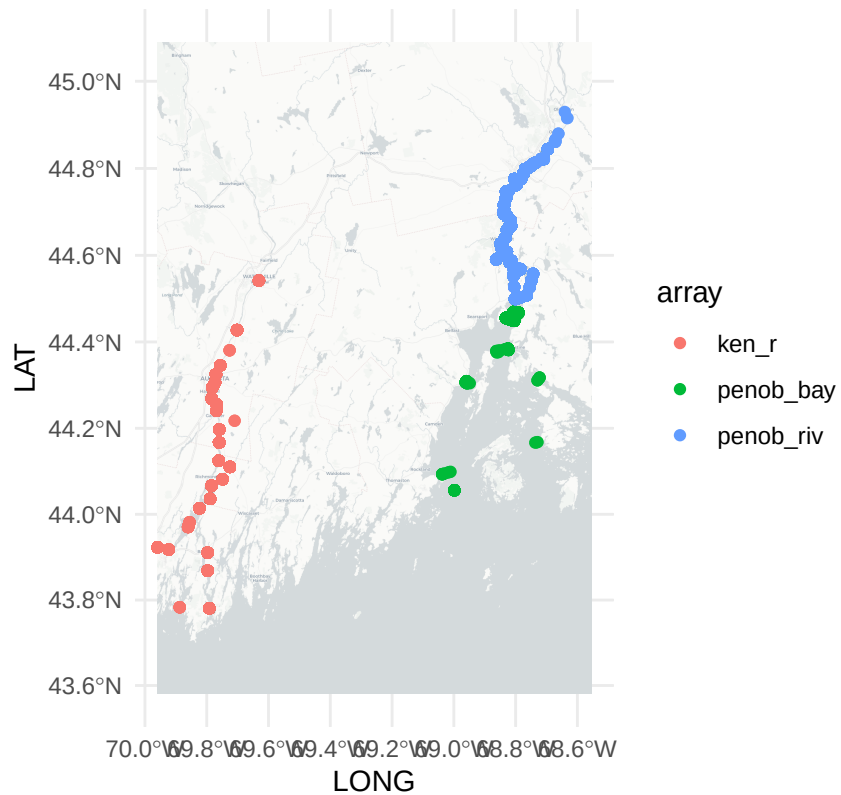
```
bb <- st_bbox(detglat_sf)
osm <- osm.raster(bb, type = "cartolight")
```

```
## Zoom: 10
```

```
osm_df <- as.data.frame(osm, xy = TRUE)

ggplot() +
  geom_raster(data = osm_df,
    aes(x = x, y = y,
      fill = rgb(layer.1/255, layer.2/255, layer.3/255))) +
  scale_fill_identity() +
  geom_sf(data = detglat_sf_3857a, aes(color = array)) +
  coord_sf(crs = 3857) +
  theme_minimal() +
  labs(title = "Detection Locations",
    x = "LONG",
    y = "LAT")
```

Detection Locations



Get Penobscot Departures

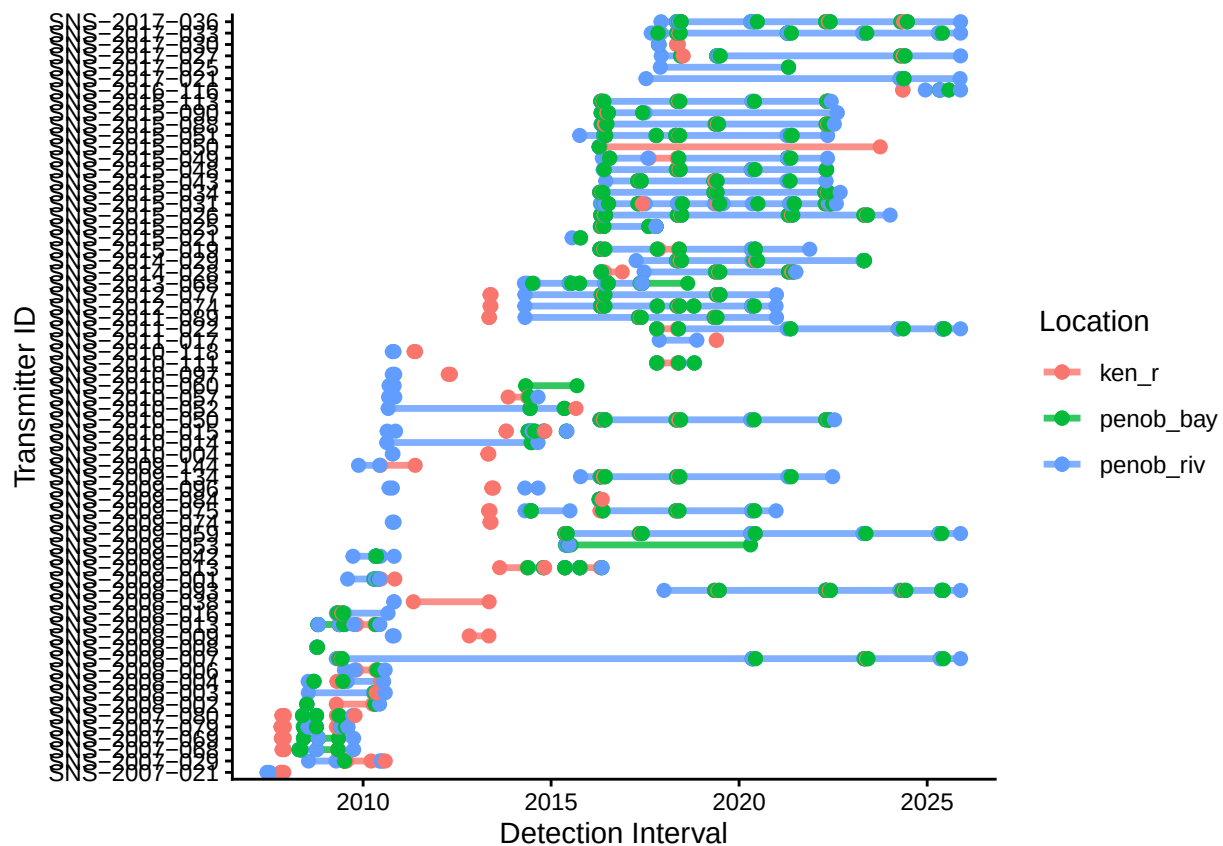
```
penob_events <- glat_events %>%
  arrange(animal_id, last_detection) %>%
  group_by(animal_id) %>%
  mutate(is_last = row_number() == n()) %>% # TRUE only for final row
  filter(!(is_last & location == "penob_riv")) %>% # drop only final penob_riv
  dplyr::select(-is_last) %>%
  ungroup() %>%
  filter(location == "penob_riv")
```

Plot Events

```
penob_cyc <- glat_events %>%
  filter(animal_id %in% penob_events$animal_id)

ggplot(penob_cyc, aes(
  y = animal_id,
  x = first_detection,
  xend = last_detection,
  yend = animal_id,
  color = location)) +
```

```
geom_segment(linewidth = 1.2, alpha = 0.8) +
labs(
  x = "Detection Interval",
  y = "Transmitter ID",
  color = "Location") +
geom_point(aes(x = first_detection), size = 2) +
geom_point(aes(x = last_detection), size = 2) +
theme_classic()
```



Fish

Keep important data

```
SNSfshc1n <- SNSfsh %>%
  dplyr::select(FishID, TagID, CaptureDate, ForkLength, TotalLength, Mass, Recapture, InitialRelease, S
```

Prep fish for Join

```
departfish <- SNSfshc1n %>%
  filter(FishID %in% penob_events$animal_id & InitialRelease == 1)%>%
  rename(animal_id = FishID)
```


Seasons

Join fish and detections

```
depart_full <- penob_events %>%
  left_join(departfish, by = "animal_id") %>%
  mutate(CaptureDate = as.POSIXct(CaptureDate, format = "%Y-%m-%d %H:%M:%S"),
         day_cap = as.numeric(difftime(last_detection, CaptureDate, units = "days")),
         day_syst = res_time_sec/86400) %>%
  filter(day_syst > 2)
glimpse(depart_full)
```

```
## Rows: 154
## Columns: 24
## $ event      <dbl> 37, 39, 44, 49, 53, 60, 68, 81, 85, 95, 99, 103, 106, ~
## $ animal_id  <chr> "SNS-2007-021", "SNS-2007-029", "SNS-2007-029", "SNS-2~
## $ location    <chr> "penob_riv", "penob_riv", "penob_riv", "penob_riv", "p~
## $ mean_latitude <dbl> 44.61727, 44.59673, 44.66246, 44.54432, 44.56275, 44.5~
## $ mean_longitude <dbl> -68.84647, -68.81292, -68.82811, -68.80062, -68.78724,~
## $ first_detection <dtm> 2007-06-13, 2008-07-20, 2010-06-08, 2008-04-19, 2008--
## $ last_detection <dtm> 2007-07-07, 2009-04-14, 2010-06-28, 2008-05-04, 2008--
## $ num_detections <int> 15, 21, 91, 36, 21, 54, 19, 18, 25, 99, 44, 207, 18, 2~
## $ res_time_sec  <dbl> 2073600, 23155200, 1728000, 1296000, 13392000, 1235520~
## $ m_leave       <dbl> 7, 4, 6, 5, 10, 10, 7, 10, 9, 4, 9, 6, 10, 5, 5, 5, 5,~
## $ y_leave       <dbl> 2007, 2009, 2010, 2008, 2008, 2008, 2008, 2008, 2009, ~
## $ d_leave       <dbl> 188, 104, 179, 125, 282, 298, 195, 279, 266, 102, 254,~
## $ y_arrive      <dbl> 2007, 2008, 2010, 2008, 2008, 2008, 2008, 2008, 2009, ~
## $ d_arrive      <dbl> 164, 202, 159, 110, 127, 155, 155, 145, 131, 200, 199,~
## $ TagID         <chr> "1039844", "1042795", "1042795", "1043689", "1043689",~
## $ CaptureDate   <dtm> 2007-05-30 09:35:00, 2007-05-30 11:30:00, 2007-05-30 ~
## $ ForkLength    <dbl> 92, 83, 83, 88, 88, 87, 94, 87, 87, 80, 86, 86, 95, 98~
## $ TotalLength   <dbl> 104, 96, 96, 100, 100, 99, 108, 100, 100, 88, 97, 97, ~
## $ Mass          <dbl> 7.75, 4.50, 4.50, 5.50, 5.50, 7.00, 9.50, 6.50, 6.50, ~
## $ Recapture     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ InitialRelease <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ Sex           <chr> "U", "U", "U", "U", "U", "F", "F", "F", "F", "U", "F",~
## $ day_cap       <dbl> 37.60069, 684.52083, 1124.52083, 221.45069, 378.45069,~
## $ day_syst      <dbl> 24, 268, 20, 15, 155, 143, 40, 134, 135, 633, 55, 328,~
```

Create Seasons

```
depart_ssn <- depart_full %>%
  mutate(season = case_when(d_leave > 240 ~ "fall",
                           d_leave < 136 ~ "spring",
                           d_leave > 136 & d_leave < 240 ~ "summer"),
         arrive_ssn = case_when(d_arrive >= 240 ~ "fall",
                                d_arrive <= 136 ~ "spring",
                                d_arrive > 136 & d_arrive < 240 ~ "summer")) %>%
  rename(year = y_leave)
glimpse(depart_ssn)
```

```
## Rows: 154
## Columns: 26
## $ event      <dbl> 37, 39, 44, 49, 53, 60, 68, 81, 85, 95, 99, 103, 106, ~
## $ animal_id  <chr> "SNS-2007-021", "SNS-2007-029", "SNS-2007-029", "SNS-2~
## $ location   <chr> "penob_riv", "penob_riv", "penob_riv", "penob_riv", "p~
## $ mean_latitude <dbl> 44.61727, 44.59673, 44.66246, 44.54432, 44.56275, 44.5~
## $ mean_longitude <dbl> -68.84647, -68.81292, -68.82811, -68.80062, -68.78724,~
## $ first_detection <dtm> 2007-06-13, 2008-07-20, 2010-06-08, 2008-04-19, 2008--
## $ last_detection <dtm> 2007-07-07, 2009-04-14, 2010-06-28, 2008-05-04, 2008--
## $ num_detections <int> 15, 21, 91, 36, 21, 54, 19, 18, 25, 99, 44, 207, 18, 2~
## $ res_time_sec <dbl> 2073600, 23155200, 1728000, 1296000, 13392000, 1235520~
## $ m_leave     <dbl> 7, 4, 6, 5, 10, 10, 7, 10, 9, 4, 9, 6, 10, 5, 5, 5, 5,~
## $ year        <dbl> 2007, 2009, 2010, 2008, 2008, 2008, 2008, 2008, 2009, ~
## $ d_leave     <dbl> 188, 104, 179, 125, 282, 298, 195, 279, 266, 102, 254,~
## $ y_arrive    <dbl> 2007, 2008, 2010, 2008, 2008, 2008, 2008, 2008, 2009, ~
## $ d_arrive    <dbl> 164, 202, 159, 110, 127, 155, 155, 145, 131, 200, 199,~
## $ TagID       <chr> "1039844", "1042795", "1042795", "1043689", "1043689",~
## $ CaptureDate <dtm> 2007-05-30 09:35:00, 2007-05-30 11:30:00, 2007-05-30 ~
## $ ForkLength  <dbl> 92, 83, 83, 88, 88, 87, 94, 87, 87, 80, 86, 86, 95, 98~
## $ TotalLength <dbl> 104, 96, 96, 100, 100, 99, 108, 100, 100, 88, 97, 97, ~
## $ Mass        <dbl> 7.75, 4.50, 4.50, 5.50, 5.50, 7.00, 9.50, 6.50, 6.50, ~
## $ Recapture   <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ InitialRelease <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ Sex         <chr> "U", "U", "U", "U", "U", "F", "F", "F", "F", "U", "F",~
## $ day_cap     <dbl> 37.60069, 684.52083, 1124.52083, 221.45069, 378.45069,~
## $ day_syst    <dbl> 24, 268, 20, 15, 155, 143, 40, 134, 135, 633, 55, 328,~
## $ season      <chr> "summer", "spring", "summer", "spring", "fall", "fall"~
## $ arrive_ssn  <chr> "summer", "summer", "summer", "spring", "spring", "sum~
```

Flow

Raw USGS stream gauge

```
API_USGS_PAT = "UMOm8b4Xd4NQmdHXvgJgloj4ROEJXneryqIYXVhU"

site <- "01036390"
pcode <- "00065" # gage height

raw <- readNWISdv("01036390", "00065", "1900-01-01", Sys.Date())
```

```
## Warning in readNWISdv("01036390", "00065", "1900-01-01", Sys.Date()): NWIS
## servers are slated for decommission. Please begin to migrate to
## read_waterdata_daily.
```

```
## GET: https://waterservices.usgs.gov/nwis/dv/?site=01036390&format=waterml%2C1.1&ParameterCd=00065&St~
```

Add seasons

```

dat <- raw %>%
  rename(gage_height = X_00065_00003) %>%
  mutate(
    jday = yday(Date),
    year = year(Date),
    season = case_when(
      jday >= 87 & jday <= 136 ~ "spring",
      jday >= 137 & jday <= 217 ~ "summer",
      jday >= 254 & jday <= 321 ~ "fall",
      TRUE ~ NA_character_)
  )

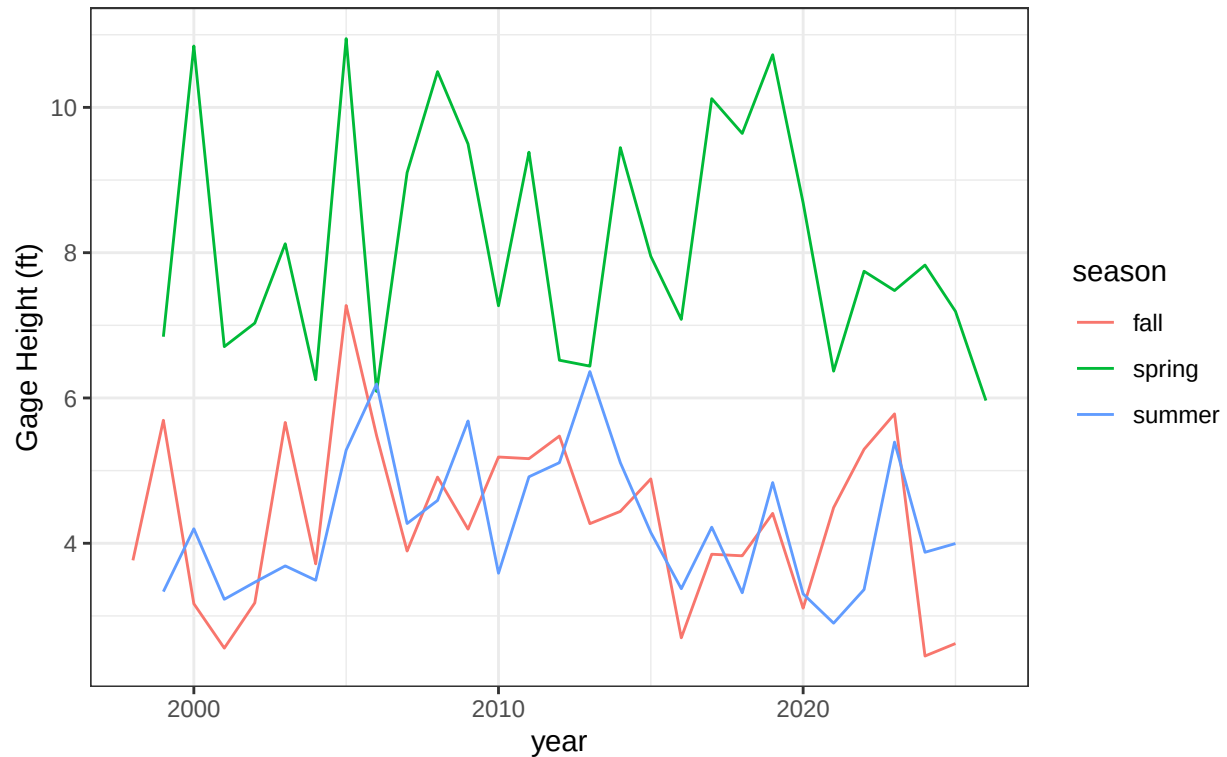
season_dat <- dat %>% filter(!is.na(season))

seasonal_means <- season_dat %>%
  group_by(year, season) %>%
  summarize(
    seasonal_mean = mean(gage_height, na.rm = TRUE),
    n_days = n(),
    .groups = "drop"
  )

ggplot(seasonal_means, aes(x = year, y = seasonal_mean, color = season)) +
  geom_line() +
  theme_bw() +
  labs(title = "Seasonal Mean Gage Height\nUSGS 01036390 Penobscot River at Eddington, ME",
    y = "Gage Height (ft)")

```

Seasonal Mean Gage Height USGS 01036390 Penobscot River at Eddington, ME



Scale flow seasons

```
flow_wide <- seasonal_means %>%
  dplyr::select(-n_days) %>%
  tidyr::pivot_wider(
    names_from = season,
    values_from = seasonal_mean,
    names_glue = "{season}_flow"
  ) %>%
  mutate(
    spring_flow_z = as.numeric(scale(spring_flow)),
    summer_flow_z = as.numeric(scale(summer_flow)),
    fall_flow_z = as.numeric(scale(fall_flow))
  )
glimpse(flow_wide)
```

```
## Rows: 29
## Columns: 7
## $ year      <dbl> 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 20~
## $ fall_flow <dbl> 3.765238, 5.693235, 3.167500, 2.556471, 3.179559, 5.6632~
## $ spring_flow <dbl> NA, 6.843400, 10.844600, 6.706600, 7.030400, 8.122800, 6~
## $ summer_flow <dbl> NA, 3.334691, 4.199383, 3.228765, 3.463704, 3.688765, 3.~
## $ spring_flow_z <dbl> NA, -0.804587935, 1.688861982, -0.889838347, -0.68805411~
## $ summer_flow_z <dbl> NA, -0.96713973, -0.07137066, -1.07687272, -0.83349057, ~
```

```
## $ fall_flow_z    <dbl> -0.48263588, 1.14404091, -0.98695543, -1.50248904, -0.97~
```

```
depart_flow <- depart_ssn %>%
  left_join(flow_wide %>%
    dplyr::select(year, spring_flow_z, summer_flow_z, fall_flow_z), by = "year")

glimpse(depart_flow)
```

```
## Rows: 154
## Columns: 29
## $ event          <dbl> 37, 39, 44, 49, 53, 60, 68, 81, 85, 95, 99, 103, 106, ~
## $ animal_id      <chr> "SNS-2007-021", "SNS-2007-029", "SNS-2007-029", "SNS-2~
## $ location       <chr> "penob_riv", "penob_riv", "penob_riv", "penob_riv", "p~
## $ mean_latitude  <dbl> 44.61727, 44.59673, 44.66246, 44.54432, 44.56275, 44.5~
## $ mean_longitude <dbl> -68.84647, -68.81292, -68.82811, -68.80062, -68.78724,~
## $ first_detection <dtm> 2007-06-13, 2008-07-20, 2010-06-08, 2008-04-19, 2008-~
## $ last_detection  <dtm> 2007-07-07, 2009-04-14, 2010-06-28, 2008-05-04, 2008-~
## $ num_detections <int> 15, 21, 91, 36, 21, 54, 19, 18, 25, 99, 44, 207, 18, 2~
## $ res_time_sec    <dbl> 2073600, 23155200, 1728000, 1296000, 13392000, 1235520~
## $ m_leave         <dbl> 7, 4, 6, 5, 10, 10, 7, 10, 9, 4, 9, 6, 10, 5, 5, 5, 5,~
## $ year            <dbl> 2007, 2009, 2010, 2008, 2008, 2008, 2008, 2008, 2009, ~
## $ d_leave         <dbl> 188, 104, 179, 125, 282, 298, 195, 279, 266, 102, 254,~
## $ y_arrive        <dbl> 2007, 2008, 2010, 2008, 2008, 2008, 2008, 2008, 2009, ~
## $ d_arrive        <dbl> 164, 202, 159, 110, 127, 155, 155, 145, 131, 200, 199,~
## $ TagID           <chr> "1039844", "1042795", "1042795", "1043689", "1043689",~
## $ CaptureDate     <dtm> 2007-05-30 09:35:00, 2007-05-30 11:30:00, 2007-05-30 ~
## $ ForkLength      <dbl> 92, 83, 83, 88, 88, 87, 94, 87, 87, 80, 86, 86, 95, 98~
## $ TotalLength     <dbl> 104, 96, 96, 100, 100, 99, 108, 100, 100, 88, 97, 97, ~
## $ Mass            <dbl> 7.75, 4.50, 4.50, 5.50, 5.50, 7.00, 9.50, 6.50, 6.50, ~
## $ Recapture       <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ InitialRelease  <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ Sex             <chr> "U", "U", "U", "U", "U", "F", "F", "F", "F", "U", "F",~
## $ day_cap         <dbl> 37.60069, 684.52083, 1124.52083, 221.45069, 378.45069,~
## $ day_syst        <dbl> 24, 268, 20, 15, 155, 143, 40, 134, 135, 633, 55, 328,~
## $ season          <chr> "summer", "spring", "summer", "spring", "fall", "fall"~
## $ arrive_ssn      <chr> "summer", "summer", "summer", "spring", "spring", "sum~
## $ spring_flow_z   <dbl> 0.6029163, 0.8485721, -0.5392398, 1.4692549, 1.4692549~
## $ summer_flow_z   <dbl> 0.00383097, 1.46617013, -0.70508507, 0.33251836, 0.332~
## $ fall_flow_z     <dbl> -0.3745189, -0.1202882, 0.7172213, 0.4842075, 0.484207~
```

Final Dataset

```
snsdep <- depart_flow %>%
  filter(!is.na(Mass))
glimpse(snsdep)
```

```
## Rows: 148
## Columns: 29
## $ event          <dbl> 37, 39, 44, 49, 53, 60, 68, 81, 85, 95, 99, 103, 106, ~
## $ animal_id      <chr> "SNS-2007-021", "SNS-2007-029", "SNS-2007-029", "SNS-2~
## $ location       <chr> "penob_riv", "penob_riv", "penob_riv", "penob_riv", "p~
```

```
## $ mean_latitude <dbl> 44.61727, 44.59673, 44.66246, 44.54432, 44.56275, 44.5~
## $ mean_longitude <dbl> -68.84647, -68.81292, -68.82811, -68.80062, -68.78724,~
## $ first_detection <dtm> 2007-06-13, 2008-07-20, 2010-06-08, 2008-04-19, 2008-~
## $ last_detection <dtm> 2007-07-07, 2009-04-14, 2010-06-28, 2008-05-04, 2008-~
## $ num_detections <int> 15, 21, 91, 36, 21, 54, 19, 18, 25, 99, 44, 207, 18, 2~
## $ res_time_sec <dbl> 2073600, 23155200, 1728000, 1296000, 13392000, 1235520~
## $ m_leave <dbl> 7, 4, 6, 5, 10, 10, 7, 10, 9, 4, 9, 6, 10, 5, 5, 5, 5,~
## $ year <dbl> 2007, 2009, 2010, 2008, 2008, 2008, 2008, 2008, 2009, ~
## $ d_leave <dbl> 188, 104, 179, 125, 282, 298, 195, 279, 266, 102, 254,~
## $ y_arrive <dbl> 2007, 2008, 2010, 2008, 2008, 2008, 2008, 2008, 2009, ~
## $ d_arrive <dbl> 164, 202, 159, 110, 127, 155, 155, 145, 131, 200, 199,~
## $ TagID <chr> "1039844", "1042795", "1042795", "1043689", "1043689",~
## $ CaptureDate <dtm> 2007-05-30 09:35:00, 2007-05-30 11:30:00, 2007-05-30 ~
## $ ForkLength <dbl> 92, 83, 83, 88, 88, 87, 94, 87, 87, 80, 86, 86, 95, 98~
## $ TotalLength <dbl> 104, 96, 96, 100, 100, 99, 108, 100, 100, 88, 97, 97, ~
## $ Mass <dbl> 7.75, 4.50, 4.50, 5.50, 5.50, 7.00, 9.50, 6.50, 6.50, ~
## $ Recapture <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ InitialRelease <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ Sex <chr> "U", "U", "U", "U", "U", "F", "F", "F", "F", "U", "F",~
## $ day_cap <dbl> 37.60069, 684.52083, 1124.52083, 221.45069, 378.45069,~
## $ day_syst <dbl> 24, 268, 20, 15, 155, 143, 40, 134, 135, 633, 55, 328,~
## $ season <chr> "summer", "spring", "summer", "spring", "fall", "fall"~
## $ arrive_ssn <chr> "summer", "summer", "summer", "spring", "spring", "sum~
## $ spring_flow_z <dbl> 0.6029163, 0.8485721, -0.5392398, 1.4692549, 1.4692549~
## $ summer_flow_z <dbl> 0.00383097, 1.46617013, -0.70508507, 0.33251836, 0.332~
## $ fall_flow_z <dbl> -0.3745189, -0.1202882, 0.7172213, 0.4842075, 0.484207~
```

Model

Order categorical variables

```
snsdep$season <- factor(snsdep$season, levels = c("spring", "summer", "fall"))
snsdep$arrive_ssn <- factor(snsdep$arrive_ssn,
                             levels = c("summer", "spring", "fall"))
```

Set Priors

```
priors <- c(
  # Fixed effects
  set_prior("normal(0, 1)", class = "b", dpar = "musummer"),
  set_prior("normal(0, 1)", class = "b", dpar = "mufall"),

  # Intercepts
  set_prior("normal(0, 1)", class = "Intercept", dpar = "musummer"),
  set_prior("normal(0, 1)", class = "Intercept", dpar = "mufall"),

  # Random effect SDs
  set_prior("exponential(1)", class = "sd", dpar = "musummer"),
```

```

  set_prior("exponential(1)", class = "sd", dpar = "mufall")
)

```

Models

```

fit_simp <- brm(
  season ~ arrive_ssn + (1 | animal_id) + (1 | year),
  data = snsdep,
  family = categorical(),
  prior = priors,    # the ones we already built
  chains = 4, cores = 4, iter = 6000,
  control = list(adapt_delta = 0.99, max_treedepth = 15)
)

fit_spring_z <- brm(
  season ~ spring_flow_z + arrive_ssn + Mass + day_cap + day_syst +
    (1 | animal_id) + (1 | year),
  data = snsdep,
  family = categorical(),
  chains = 4, iter = 6000, cores = 4,
  control = list(adapt_delta = 0.99)
)

summary(fit_spring_z)

fit_summer_z <- brm(
  season ~ summer_flow_z + arrive_ssn + Mass + day_cap + day_syst +
    (1 | animal_id) + (1 | year),
  data = snsdep,
  family = categorical()
)

summary(fit_summer_z)

fit_fall_z <- brm(
  season ~ fall_flow_z + arrive_ssn + Mass + day_cap + day_syst +
    (1 | animal_id) + (1 | year),
  data = snsdep,
  family = categorical()
)

summary(fit_fall_z)

loo(fit_spring_z, fit_summer_z, fit_fall_z)

fit_spring_summer_z <- brm(
  season ~ spring_flow_z * summer_flow_z + arrive_ssn + Mass + day_cap + day_syst +
    (1 | animal_id) + (1 | year),
  data = snsdep,

```

```

    family = categorical()
  )

fit_spring_fall_z <- brm(
  season ~ spring_flow_z * fall_flow_z + arrive_ssn + Mass + day_cap + day_syst +
    (1 | animal_id) + (1 | year),
  data = snsdep,
  family = categorical()
)

fit_summer_fall_z <- brm(
  season ~ summer_flow_z * fall_flow_z + arrive_ssn + Mass + day_cap + day_syst +
    (1 | animal_id) + (1 | year),
  data = snsdep,
  family = categorical()
)

```

Compare Models

```

loo(fit_simp, fit_spring_z, fit_summer_z, fit_fall_z, fit_spring_summer_z, fit_spring_fall_z,
    fit_summer_fall_z)

```

```

## Output of model 'fit_simp':
##
## Computed from 12000 by 148 log-likelihood matrix.
##
##           Estimate   SE
## elpd_loo    -114.1  7.1
## p_loo         37.8  3.1
## looic        228.2 14.2
## -----
## MCSE of elpd_loo is NA.
## MCSE and ESS estimates assume MCMC draws (r_eff in [0.3, 1.4]).
##
## Pareto k diagnostic values:
##               Count Pct.   Min. ESS
## (-Inf, 0.7]  (good)   146  98.6%   806
## (0.7, 1]     (bad)     2   1.4%   <NA>
## (1, Inf)     (very bad) 0   0.0%   <NA>
## See help('pareto-k-diagnostic') for details.
##
## Output of model 'fit_spring_z':
##
## Computed from 12000 by 148 log-likelihood matrix.
##
##           Estimate   SE
## elpd_loo    -110.1 11.8
## p_loo         73.3  8.6
## looic        220.1 23.5
## -----

```



```

## MCSE of elpd_loo is NA.
## MCSE and ESS estimates assume MCMC draws (r_eff in [0.6, 1.2]).
##
## Pareto k diagnostic values:
##               Count Pct.   Min. ESS
## (-Inf, 0.7]   (good)    80   54.1%   330
##   (0.7, 1]    (bad)     61   41.2%   <NA>
##   (1, Inf)    (very bad) 7    4.7%   <NA>
## See help('pareto-k-diagnostic') for details.
##
## Output of model 'fit_summer_z':
##
## Computed from 4000 by 148 log-likelihood matrix.
##
##           Estimate   SE
## elpd_loo   -103.7 11.3
## p_loo       67.8  8.2
## looic       207.3 22.7
## -----
## MCSE of elpd_loo is NA.
## MCSE and ESS estimates assume MCMC draws (r_eff in [0.5, 1.3]).
##
## Pareto k diagnostic values:
##               Count Pct.   Min. ESS
## (-Inf, 0.7]   (good)    83   56.1%   91
##   (0.7, 1]    (bad)     55   37.2%   <NA>
##   (1, Inf)    (very bad) 10    6.8%   <NA>
## See help('pareto-k-diagnostic') for details.
##
## Output of model 'fit_fall_z':
##
## Computed from 4000 by 148 log-likelihood matrix.
##
##           Estimate   SE
## elpd_loo   -106.7 12.0
## p_loo       70.9  8.9
## looic       213.5 24.0
## -----
## MCSE of elpd_loo is NA.
## MCSE and ESS estimates assume MCMC draws (r_eff in [0.5, 1.3]).
##
## Pareto k diagnostic values:
##               Count Pct.   Min. ESS
## (-Inf, 0.7]   (good)    80   54.1%   248
##   (0.7, 1]    (bad)     55   37.2%   <NA>
##   (1, Inf)    (very bad) 13    8.8%   <NA>
## See help('pareto-k-diagnostic') for details.
##
## Output of model 'fit_spring_summer_z':
##
## Computed from 4000 by 148 log-likelihood matrix.
##
##           Estimate   SE
## elpd_loo   -106.0 12.3

```

```

## p_loo          74.7  9.3
## looic          212.0 24.6
## -----
## MCSE of elpd_loo is NA.
## MCSE and ESS estimates assume MCMC draws (r_eff in [0.5, 1.3]).
##
## Pareto k diagnostic values:
##               Count Pct.   Min. ESS
## (-Inf, 0.7] (good)    57  38.5%   218
##   (0.7, 1] (bad)     73  49.3%   <NA>
##   (1, Inf) (very bad) 18  12.2%   <NA>
## See help('pareto-k-diagnostic') for details.
##
## Output of model 'fit_spring_fall_z':
##
## Computed from 4000 by 148 log-likelihood matrix.
##
##           Estimate   SE
## elpd_loo  -103.6 12.0
## p_loo      73.1  9.0
## looic      207.2 24.1
## -----
## MCSE of elpd_loo is NA.
## MCSE and ESS estimates assume MCMC draws (r_eff in [0.5, 1.2]).
##
## Pareto k diagnostic values:
##               Count Pct.   Min. ESS
## (-Inf, 0.7] (good)    55  37.2%   182
##   (0.7, 1] (bad)     70  47.3%   <NA>
##   (1, Inf) (very bad) 23  15.5%   <NA>
## See help('pareto-k-diagnostic') for details.
##
## Output of model 'fit_summer_fall_z':
##
## Computed from 4000 by 148 log-likelihood matrix.
##
##           Estimate   SE
## elpd_loo  -104.6 12.4
## p_loo      74.3  9.5
## looic      209.2 24.8
## -----
## MCSE of elpd_loo is NA.
## MCSE and ESS estimates assume MCMC draws (r_eff in [0.5, 1.3]).
##
## Pareto k diagnostic values:
##               Count Pct.   Min. ESS
## (-Inf, 0.7] (good)    51  34.5%   142
##   (0.7, 1] (bad)     77  52.0%   <NA>
##   (1, Inf) (very bad) 20  13.5%   <NA>
## See help('pareto-k-diagnostic') for details.
##
## Model comparisons:
##               elpd_diff se_diff
## fit_spring_fall_z      0.0      0.0

```

```
## fit_summer_z          -0.1      3.1
## fit_summer_fall_z     -1.0      3.5
## fit_spring_summer_z   -2.4      2.4
## fit_fall_z            -3.1      3.1
## fit_spring_z          -6.5      2.8
## fit_simp              -10.5     8.6
```

Results Figures

Summary of two best models

```
summary(fit_simp) ## best overall
```

```
## Family: categorical
## Links: musummer = logit; mufall = logit
## Formula: season ~ arrive_ssn + (1 | animal_id) + (1 | year)
## Data: snsdep (Number of observations: 148)
## Draws: 4 chains, each with iter = 6000; warmup = 3000; thin = 1;
## total post-warmup draws = 12000
##
## Multilevel Hyperparameters:
## ~animal_id (Number of levels: 60)
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS
## sd(musummer_Intercept)    0.97    0.60    0.05    2.29 1.00    1948
## sd(mufall_Intercept)      1.10    0.77    0.04    2.88 1.01    1644
##           Tail_ESS
## sd(musummer_Intercept)    3239
## sd(mufall_Intercept)      3640
##
## ~year (Number of levels: 16)
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS
## sd(musummer_Intercept)    1.53    0.61    0.49    2.91 1.00    2915
## sd(mufall_Intercept)      2.28    0.82    1.02    4.18 1.00    3909
##           Tail_ESS
## sd(musummer_Intercept)    2430
## sd(mufall_Intercept)      5348
##
## Regression Coefficients:
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS
## musummer_Intercept        -0.99    0.49   -1.96   -0.02 1.00    6180
## mufall_Intercept          -1.46    0.61   -2.68   -0.26 1.00    6399
## musummer_arrive_ssnspring -1.29    0.64   -2.59   -0.07 1.00   12290
## musummer_arrive_ssnfall   -0.01    0.63   -1.26    1.22 1.00   12734
## mufall_arrive_ssnspring   -0.82    0.69   -2.19    0.52 1.00   12560
## mufall_arrive_ssnfall      0.98    0.64   -0.29    2.21 1.00   11737
##           Tail_ESS
## musummer_Intercept        7678
## mufall_Intercept          7379
## musummer_arrive_ssnspring  9261
## musummer_arrive_ssnfall    9363
## mufall_arrive_ssnspring    8544
```

```
## mufall_arrive_ssnfall          7984
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

summary(fit_summer_z) ## best with variables

## Family: categorical
## Links: musummer = logit; mufall = logit
## Formula: season ~ summer_flow_z + arrive_ssn + Mass + day_cap + day_syst + (1 | animal_id) + (1 | year)
## Data: snsdep (Number of observations: 148)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
## total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~animal_id (Number of levels: 60)
##
```

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS
sd(musummer_Intercept)	3.43	1.42	1.19	6.69	1.00	646
sd(mufall_Intercept)	5.39	2.57	1.83	11.70	1.01	463
	Tail_ESS					
sd(musummer_Intercept)	1171					
sd(mufall_Intercept)	665					

```
##
## ~year (Number of levels: 16)
##
```

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS
sd(musummer_Intercept)	3.05	1.43	0.92	6.52	1.00	835
sd(mufall_Intercept)	2.31	1.64	0.14	6.27	1.00	804
	Tail_ESS					
sd(musummer_Intercept)	1089					
sd(mufall_Intercept)	1206					

```
##
## Regression Coefficients:
##
```

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS
musummer_Intercept	0.12	3.11	-6.61	5.85	1.00	1877
mufall_Intercept	7.02	5.18	-1.26	18.78	1.01	889
musummer_summer_flow_z	0.80	1.32	-1.95	3.43	1.00	1647
musummer_arrive_ssnspring	-7.01	2.74	-13.29	-2.80	1.00	871
musummer_arrive_ssnfall	-1.09	1.67	-4.40	2.13	1.00	2098
musummer_Mass	0.38	0.46	-0.43	1.39	1.00	1355
musummer_day_cap	-0.00	0.00	-0.00	0.00	1.00	3085
musummer_day_syst	-0.01	0.00	-0.01	-0.00	1.00	1144
mufall_summer_flow_z	2.58	1.63	-0.28	6.16	1.01	889
mufall_arrive_ssnspring	-7.23	3.82	-16.06	-2.02	1.01	758
mufall_arrive_ssnfall	-0.68	2.09	-5.26	2.95	1.00	1365
mufall_Mass	-0.44	0.67	-1.93	0.75	1.00	1329
mufall_day_cap	-0.00	0.00	-0.00	0.00	1.01	953
mufall_day_syst	-0.01	0.00	-0.02	-0.01	1.01	875
	Tail_ESS					
musummer_Intercept	1873					
mufall_Intercept	1270					
musummer_summer_flow_z	1913					
musummer_arrive_ssnspring	1206					
musummer_arrive_ssnfall	1807					

```
## musummer_Mass          1760
## musummer_day_cap      2592
## musummer_day_syst     2539
## mufall_summer_flow_z   1114
## mufall_arrive_ssnspring 722
## mufall_arrive_ssnfall  1389
## mufall_Mass           1554
## mufall_day_cap        968
## mufall_day_syst       595
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

Arrival season (Figure 1)

```
coefs <- summary(fit_simp)$fixed

# Keep only arrival-season effects
arrive_coefs <- coefs[grep("arrive_ssn", rownames(coefs)), ]
nice_names <- c(
  "musummer_arrive_ssnspring" = "Arrive Spring → Depart Summer",
  "musummer_arrive_ssnfall"   = "Arrive Fall → Depart Summer",
  "mufall_arrive_ssnspring"   = "Arrive Spring → Depart Fall",
  "mufall_arrive_ssnfall"     = "Arrive Fall → Depart Fall"
)

rownames(arrive_coefs) <- nice_names[rownames(arrive_coefs)]

par(mar = c(6, 15, 4, 2))

# Reverse order so the first coefficient appears at the top
k <- nrow(arrive_coefs)
y <- k:1

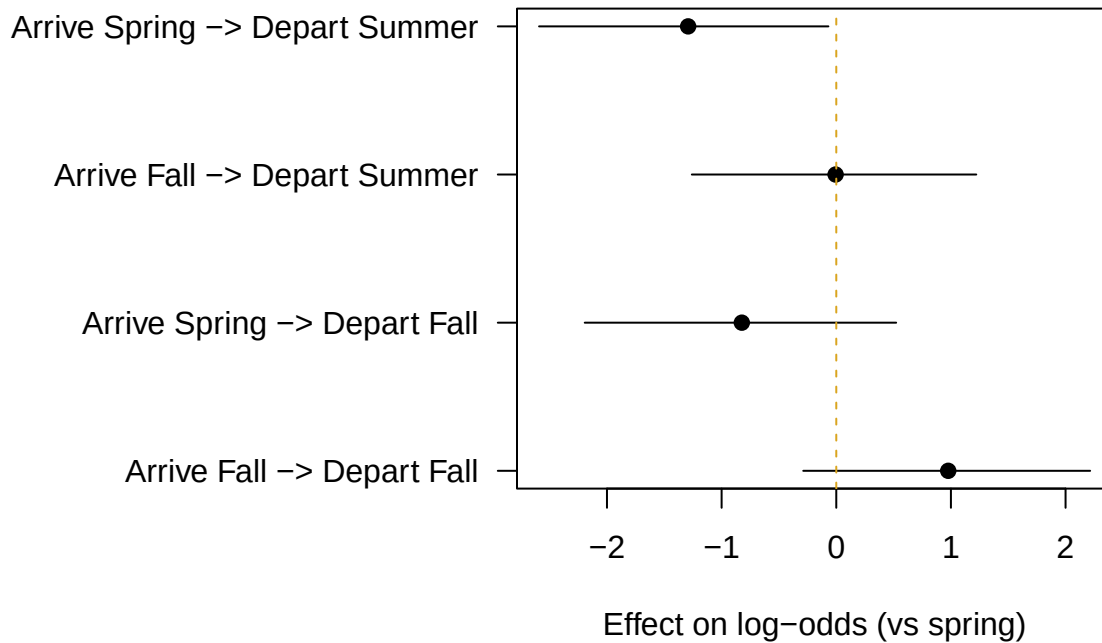
plot(arrive_coefs[, "Estimate"], y,
     xlab = "Effect on log-odds (vs spring)",
     ylab = "",
     yaxt = "n",
     pch = 19,
     xlim = range(c(arrive_coefs[, "l-95% CI"], arrive_coefs[, "u-95% CI"])),
     main = "Arrival-season effects on departure timing")

segments(arrive_coefs[, "l-95% CI"], y,
         arrive_coefs[, "u-95% CI"], y)

axis(2, at = y, labels = rownames(arrive_coefs), las = 1)

abline(v = 0, col = "goldenrod", lty = 2)
```

Arrival-season effects on departure timing



Flow anomalies (Figure 2)

```

coefs <- summary(fit_summer_z)$fixed

flow_coefs <- coefs[grepl("summer_flow_z", rownames(coefs)), ]
flow_coefs

##              Estimate Est.Error  1-95% CI u-95% CI      Rhat
## musummer_summer_flow_z 0.7974494  1.318769 -1.9455841 3.432545 1.000882
## mufall_summer_flow_z   2.5834933  1.632151 -0.2829687 6.158258 1.005945
##              Bulk_ESS Tail_ESS
## musummer_summer_flow_z 1647.014 1913.460
## mufall_summer_flow_z   888.745 1114.156

nice_names <- c(
  "musummer_summer_flow_z" = "Effect of flow on Summer vs Spring",
  "mufall_summer_flow_z"   = "Effect of flow on Fall vs Spring"
)

rownames(flow_coefs) <- nice_names[rownames(flow_coefs)]

par(mar = c(6, 18, 4, 2))

```

```

k <- nrow(flow_coefs)
y <- k:1

plot(flow_coefs[, "Estimate"], y,
     xlab = "Effect of summer flow anomaly (log-odds)",
     ylab = "",
     yaxt = "n",
     pch = 19,
     xlim = range(c(flow_coefs[, "l-95% CI"], flow_coefs[, "u-95% CI"])),
     main = "Flow anomaly effects on departure timing")

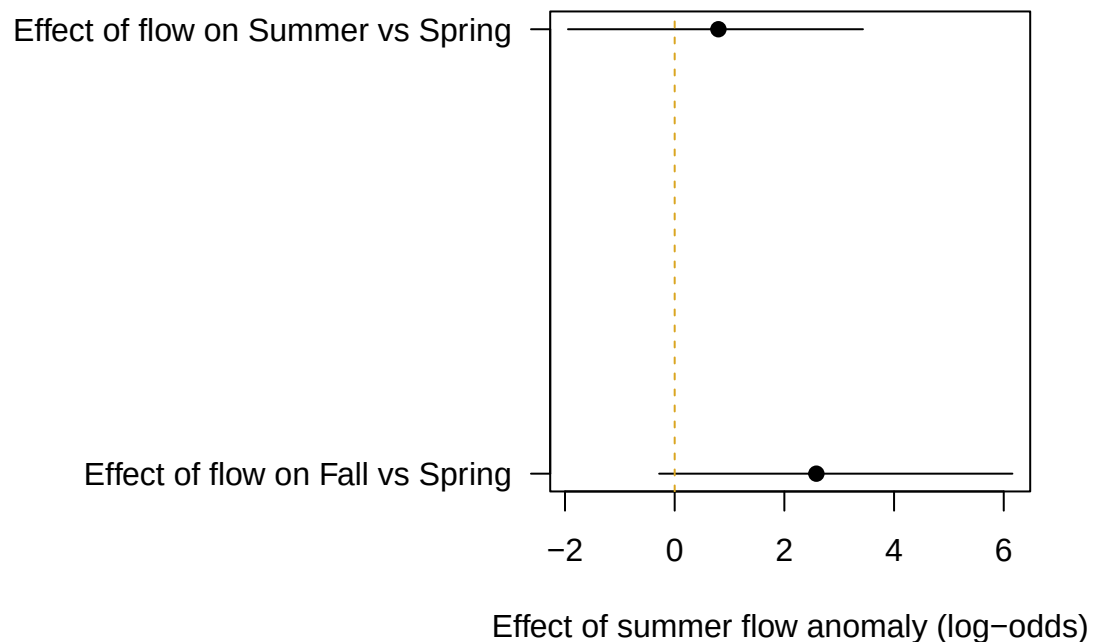
segments(flow_coefs[, "l-95% CI"], y,
         flow_coefs[, "u-95% CI"], y)

axis(2, at = y, labels = rownames(flow_coefs), las = 1)

abline(v = 0, col = "goldenrod", lty = 2)

```

Flow anomaly effects on departure timing



```

## Boxplot

#boxplot(summer_flow_z ~ season, data=snsdep,
#        xlab="Departure season",
#        ylab="Summer flow anomaly")

```

Random effects Year (Figure 3) & Individual (Figure 4)

```
year_counts <- table(snsdep$year)
id_counts <- table(snsdep$animal_id)

re <- ranef(fit_simp, summary = TRUE)

year_summer <- as.data.frame(re$year[, , "musummer_Intercept"]) %>%
  mutate(year = rownames(re$year))

year_fall <- as.data.frame(re$year[, , "mufall_Intercept"]) %>%
  mutate(year = rownames(re$year))

id_summer <- as.data.frame(re$animal_id[, , "musummer_Intercept"]) %>%
  mutate(animal_id = rownames(re$animal_id))

id_fall <- as.data.frame(re$animal_id[, , "mufall_Intercept"]) %>%
  mutate(animal_id = rownames(re$animal_id))

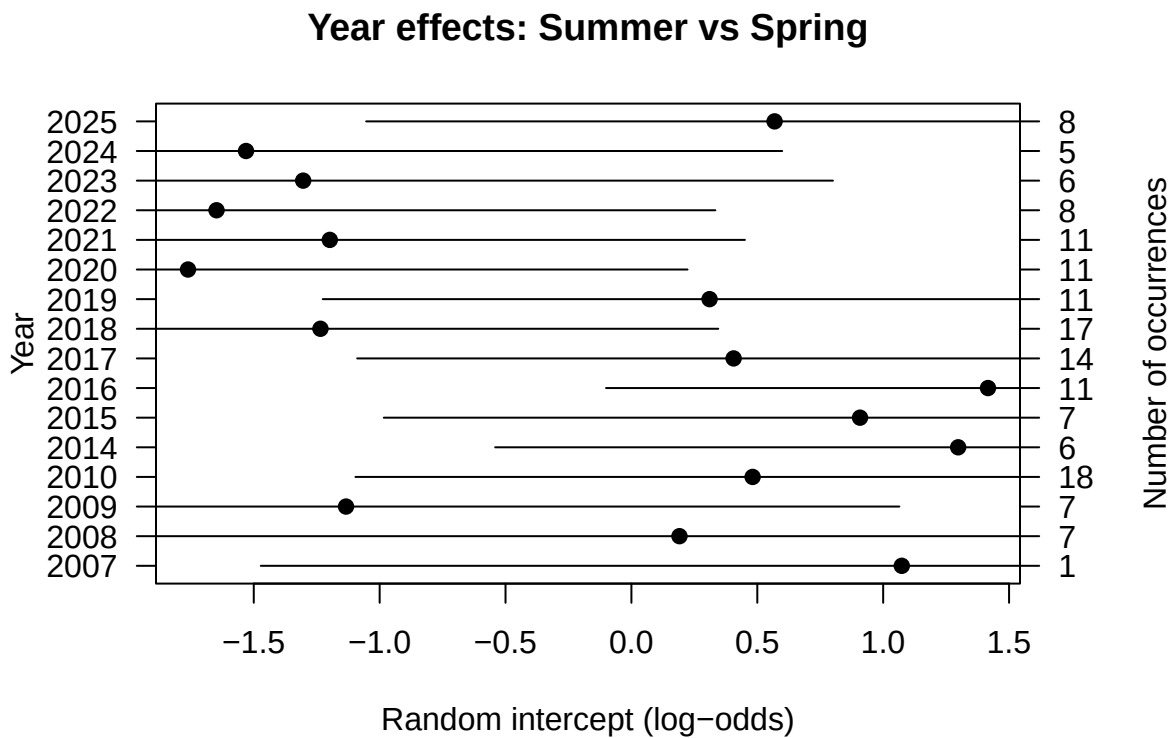
## Year

### Summer vs Spring

par(mar = c(6, 5, 4, 5))

with(year_summer, {
  plot(Estimate, seq_along(year),
       xlab = "Random intercept (log-odds)",
       ylab = "Year",
       main = "Year effects: Summer vs Spring",
       pch = 19, yaxt = "n")
  axis(2, at = seq_along(year), labels = year, las = 1)
  segments(Q2.5, seq_along(year), Q97.5, seq_along(year))

  # Secondary axis: number of occurrences
  axis(4, at = seq_along(year), labels = year_counts[year], las = 1)
  mtext("Number of occurrences", side = 4, line = 3)
})
```

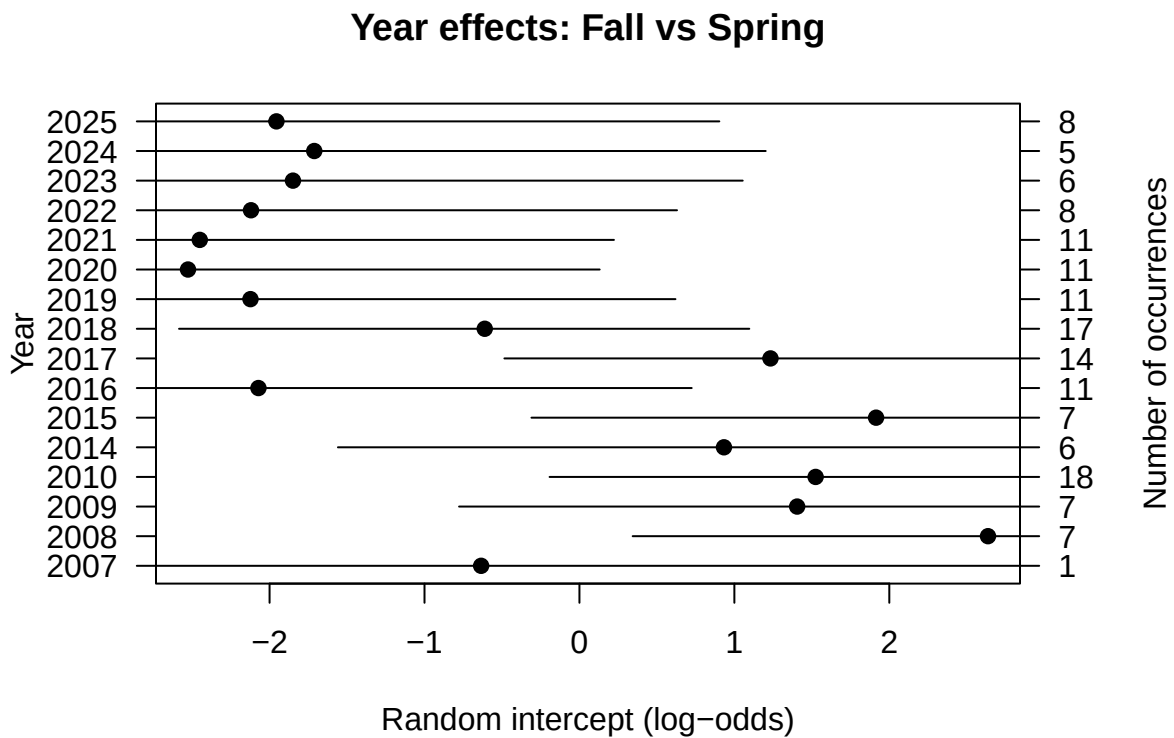
```

### Fall vs Spring
par(mar = c(6, 5, 4, 5))

with(year_fall, {
  plot(Estimate, seq_along(year),
       xlab = "Random intercept (log-odds)",
       ylab = "Year",
       main = "Year effects: Fall vs Spring",
       pch = 19, yaxt = "n")
  axis(2, at = seq_along(year), labels = year, las = 1)
  segments(Q2.5, seq_along(year), Q97.5, seq_along(year))

  # Secondary axis
  axis(4, at = seq_along(year), labels = year_counts[year], las = 1)
  mtext("Number of occurrences", side = 4, line = 3)
})

```



```

### Individual

### Summer vs Spring

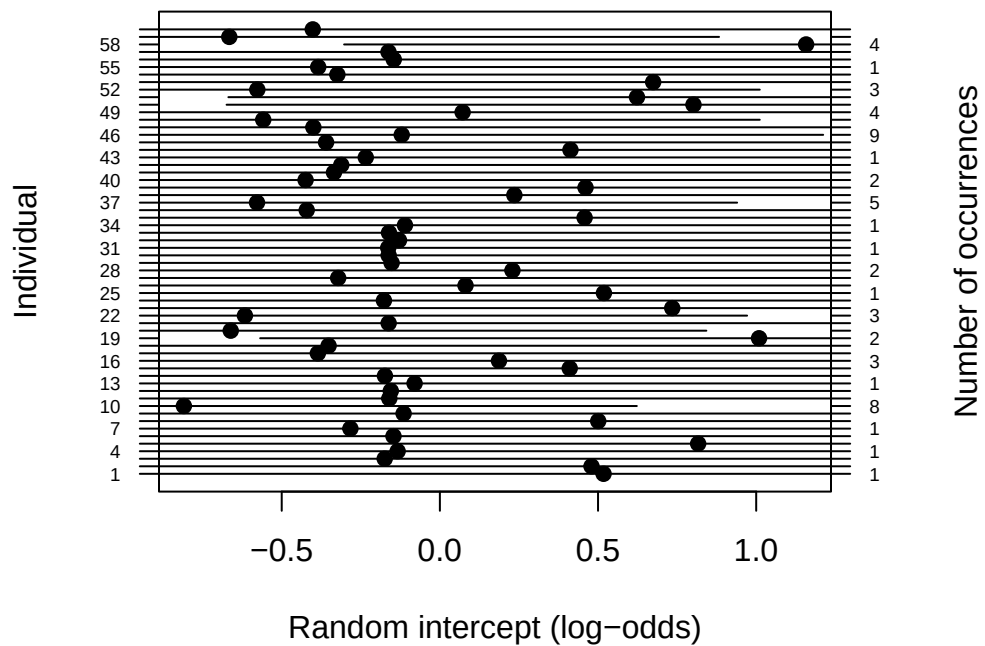
par(mar = c(6, 10, 4, 5))

with(id_summer, {
  plot(Estimate, seq_along(animal_id),
       xlab = "Random intercept (log-odds)",
       ylab = "Individual",
       main = "Individual effects: Summer vs Spring",
       pch = 19, yaxt = "n")
  axis(2, at = seq_along(animal_id), las = 1, cex.axis = 0.6)
  segments(Q2.5, seq_along(animal_id), Q97.5, seq_along(animal_id))

  # Secondary axis
  axis(4, at = seq_along(animal_id), labels = id_counts[animal_id], las = 1, cex.axis = 0.6)
  mtext("Number of occurrences", side = 4, line = 3)
})

```

Individual effects: Summer vs Spring

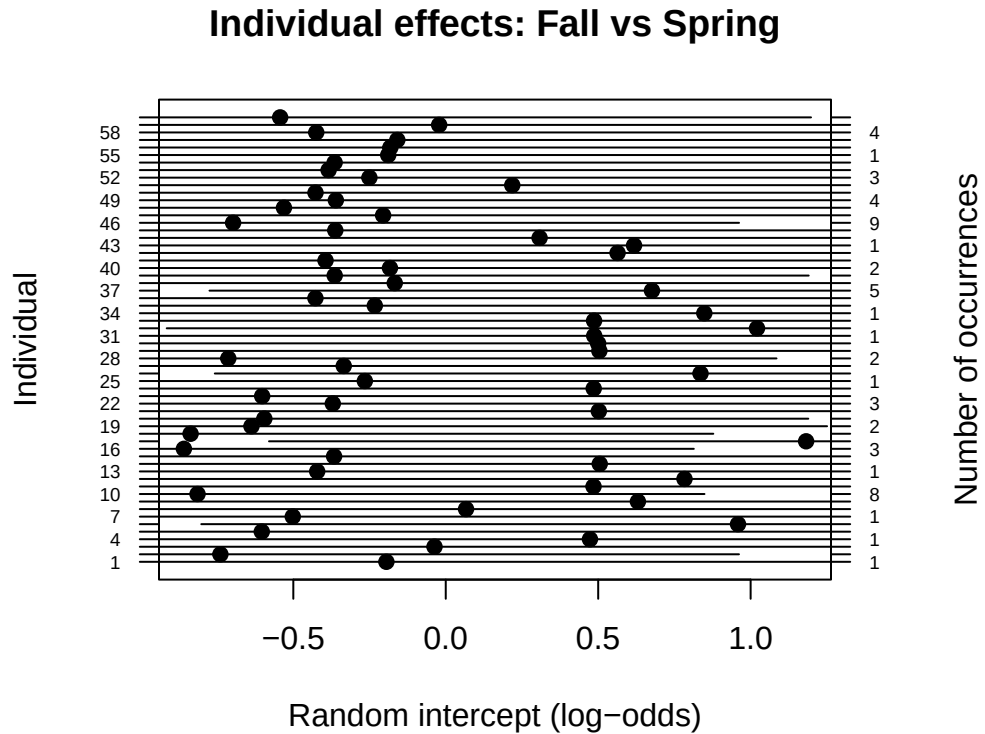


```
### Fall vs Spring

par(mar = c(6, 10, 4, 5))

with(id_fall, {
  plot(Estimate, seq_along(animal_id),
       xlab = "Random intercept (log-odds)",
       ylab = "Individual",
       main = "Individual effects: Fall vs Spring",
       pch = 19, yaxt = "n")
  axis(2, at = seq_along(animal_id), las = 1, cex.axis = 0.6)
  segments(Q2.5, seq_along(animal_id), Q97.5, seq_along(animal_id))

  # Secondary axis
  axis(4, at = seq_along(animal_id), labels = id_counts[animal_id], las = 1, cex.axis = 0.6)
  mtext("Number of occurrences", side = 4, line = 3)
})
```



Results Writing

The strongest model included only arrival season for fixed effect, with individual and year as random effects. This model was close too the rest in elpd (> 9 from the top model), but it had far better Pareto-k values (98.6% vs 78% good). It is also true that the simpler model is better when elpd is close. The simple model showed that arrival season predicted departure season. Fish that arrived in the spring were less likely to depart in the summer or fall. Fish arriving in the fall were more likely to depart in the fall (Figure 1). Many fish stayed for one or more years, so this is not a result of fish arriving and leaving right away. Year to year variation was significant and the largest of the effects. Some years strongly shifted towards spring and others fall, even in years with up to 17 departures (Figure 3). This model showed that individual fish was a significant factor in departure timing. Some fish are consistently more likely to depart early or late over as many as 9 occurrences. This variance is comparable in magnitude to the fixed effect.

The second best model with summer flow anomalies also included many factors with biological relevance to the fish. Fish mass, days in system, and days after capture all failed to predict departure season. Even flow was not a good predictor. Figure 2 shows that summer flow anomaly does not significantly effect departure. While it appears that higher summer flows increase fall departures relative to spring, the confidence intervals cross zero, so this cannot be said for sure. The flow effects were also much smaller than the arrival season, annual, and individual effects. This difference in broad scale year effect and short-term seasonal flow anomalies shape departure timing.